

"SELF - SUPERVISED LEARN."

→ Machine Learning Approach where the model creates its own labels from unlabeled data

→ Ex: "The cat is on the ____"

↳ the model predicts "mat"

→ Workflow of SSL

① Pretext Task

↳ Artificial task designed to train the model

↳ NLP: Predict masked word

↳ Audio: Predict missing sound

② Downstream Task

↳ After learning representation model is fine tuned for real tasks, like classification, detection etc

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→ Representation Learning

↳ The goal is to learn useful features from raw data that can be later used for downstream tasks.

→ SSL Algorithms

① Generative / Reconstruction Based Algo

↳ Learn by reconstructing corrupted or missing data

② Pretext Task Algo

↳ Learn by predicting one part of data from another part.

↳ Examples : masked word prediction

③ Clustering Based Methods

↳ Generate pseudo-labels using clustering

④ Contrastive Learning

↳ Goal :

↳ Bring similar things closer (positive pairs)

↳ Push diff things farther (neg pairs)

↳ Process :

① Data Augmentation

↳ create various transformations to unlabeled data

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↳ Increases data variability and diversification

↳ Ex: For an image: crop, rotate, jitter, blur

② Encoder Network

↳ Takes augmented instances as input and extract features from it

↳ CNN → for image, Transformer → text

③ Projection Network

↳ Projection head maps features to lower-dimensional space.

↳ Purpose:

↳ Improve discriminative power

↳ reduce redundancy

↳ improve separation

④ Model Training

↳ The objective of training is to minimize dist b/w +ve pairs & maximize dist b/w -ve pairs.

↳ Distance metric: Euclidean dist, Cosine Similarity

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→ Loss Function in Contrastive Learn.

① Contrastive Loss

↳ Goals:

↳ Similar points together

↳ Dissimilar apart

↳ for a (x_1, x_2) :

$$L = (1 - Y) \frac{1}{2} D^2 + Y \frac{1}{2} \max(0, m - D)^2$$

Where,

$Y = 0 \rightarrow +ve$ pair

$Y = 1 \rightarrow -ve$ pair

$D = \|f(x_1) - f(x_2)\|$

$m = \text{margin (min dist for } -ve)$

② Triplet Loss

↳ Uses: Anchor, +ve, -ve instances

↳ Goals:

↳ Anchor should be closer to +ve

↳ farther from -ve by at least margin

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$$L = \max(0, D(A, P) - D(A, N) + m)$$

→ SimCLR (Simplified Framework for Contra. Learn.)

↳ Learn visual representations without labels using contrastive learning

↳ NT-Xent Loss

↳ Normalized Temperature-scaled Cross Entropy loss

↳ For a batch of N images, there are $2N$ augmented samples

↳ Each +ve pair compared against $2(N-1)$ -ve pairs
↳ maximize cosine similarity for +ve pairs

$$l_{i,j} = -\log \left(\frac{\exp(\text{sim}(z_i, z_j) / \tau)}{\sum_{k=1}^{2N} 1_{[k \neq i]} \exp(\text{sim}(z_i, z_k) / \tau)} \right)$$

τ = temperature

↳ Small value pushes model to strongly separate +ve from -ves. Harder training

↳ Large value → softer separation, easier training but less precise embeddings

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→ Cross-Modal Learning (CML)

↳ Combines multiple modalities to understand relationships b/w diff data types

↳ Process:

① Unimodal Encoding

↳ Each modality processed separately

↳ Ex: Image → CNN, Text → Transformers

↳ Output → dense embedding vectors

② Representation Alignment

↳ Map all modalities into a shared embedding space

↳ Image of dog ↔ text "dog" should have nearby embeddings

③ Projection Heads

↳ Linear / Multilayer Perceptrons projections map each modalities embedding to the shared space

④ Contrastive Alignment

↳ Given batch of N (images, text) pairs, construct $N \times N$ similarity matrix

$$S_{ij} = \frac{z_i^{(img)} \cdot z_j^{(text)}}{\|z_i^{(img)}\| \cdot \|z_j^{(text)}\|}$$

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(5) Fusion is then applied, late, intermediate, early, or attention based.

Semi-Superv. Learning

- Small amount of human label
- Unlabeled data used with labeled data
- Improve supervision task

Self Supervised Learning

- No human labels
- All data is unlabeled
- Learn representations